

Physics Paper 4 Notes

Taking readings

<u>Instrument</u>	<u>Uncertainty / Precision</u>	
<u>Length</u>		
Ruler	0.1 <i>cm</i>	
Vernier Calliper	0.01 <i>cm</i>	
Micrometre Screw Gauge	0.01 <i>mm</i> / 0.001 <i>cm</i>	
<u>Volume</u>		
100 <i>cm</i> ³ Measuring cylinder	0.5 <i>cm</i> ³	
250 <i>cm</i> ³ Measuring cylinder	1 <i>cm</i> ³	
<u>Angle</u>		
Protractor	1°	
<u>Mass</u>		
Electronic Mass Balance	0.1 <i>g</i>	
Spring Balance	5 <i>g</i>	
<u>Time</u>		
Stopwatch	<i>Reaction time (0.2s)</i>	
<u>Temperature</u>		
Mercury thermometer	0.1 °C	
<u>Electricity</u>		
Analog voltmeter		
Analog ammeter		
Digital Multi-meter (4-digits display)	Ammeter	20 <i>mA</i> – 0.01 <i>mA</i>
		200 <i>mA</i> – 0.1 <i>mA</i>
	Voltmeter	
2 Times		Diameter of cylinder
		Depth of cylinder
		Time taken for n oscillations
3 Times		Diameter of sphere

Question types

1. Validity of relationship [3]

- Table ($A \propto B \Rightarrow A = kB$)

Experiment	A / units	B / units	k / units
1			
2			

Find $\frac{\Delta k}{k}$ and $\frac{\Delta A}{A} + \frac{\Delta B}{B}$

Conclusion: (compare $\frac{\Delta k}{k}$ with percentage uncertainty of A and B)

- Since percentage uncertainty of k is **less** than percentage uncertainty of $A + B$, relationship is true.
- OR
- Since percentage uncertainty of k is **more** than percentage uncertainty of $A + B$, relationship is not true.

*Include units for k !!!

2. Percentage uncertainty [1]

- **%!!!**
- **±!!!**
- Percentage uncertainty: **1 d.p**
- Account for significant human error or experimental limitations when taking the measurement
 - Time: 0.3s
 - Angle: 4°

3. Graph [6]

Linearization, gradient and y-intercept [3]:

- Plotting _____ against _____, if a straight line (that does / does NOT pass through the origin) is obtained, where _____ is the gradient, _____ is the y-intercept.

*FOR DOUBLE VARIABLE $y = ka^mb^n$,

Plotting $\ln y = m \ln a + \ln kb^n$ and $\ln y = n \ln b + \ln ka^m$

- $Gradient = \frac{y_1 - y_2}{x_1 - x_2}$
 - Gradient triangle
 - Coordinate
- To find y-intercept
 - read from graph
 - calculate

*Quoted data follow **least s. f. of table**

*If question ask for how a variable can be determined, MUST show working for the variable

E.g. $a^2 = kb^2$

Plotting a^2 against b^2 , k is the gradient of the graph.

Scale [1]:

- Data point plotted must cover (≥ 6 big square) $\times$ (≥ 4 big square)

Best fit line [1]:

- *Distance of data point to line ≤ 2 mini square*
- ≤ 1 anomalous point
- **Circle & label** anomalous point
 - There are no anomalous points as all plotted points follow the trend of the best fit line.
 - There is one anomalous point as it lies relatively far from the best fit line compared to the other points / it does not follow the trend of the best fit line.
- Every big square must be labelled

4. Source of error & Improvement

- Error + How it affects the experiment

Source of error	Improvement
Unstable measurement – When measuring θ / length, protractor / ruler comes into contact with the set-up, causing disturbances and θ / length cannot be measured accurately.	Secure protractor / ruler with a retort stand such that the measurement is reliable.
It is difficult to read θ as the centre line of the spring is hard to determine / rod is thick hence affecting the value of θ measured.	Make a reference line at the centre of the rod with a marker.
There is no vertical reference to use in measuring θ , affecting the value of θ measured.	
The number of readings taken is insufficient to plot the graph.	Take more readings.
Friction present in the ball bearings of the pulley causes the setup to be stable over a range of θ	Lubricate the axle of the pulley
The temperature of the thermometer does not reflect the actual temperature of the thermistor	Stir to ensure thermodynamic equilibrium is reached.
For PROJECTILE experiments: Difficult to gauge where the vertical axis is	Use a plumb line to indicate the vertical axis
For experiments requiring 2 measurements to be taken simultaneously (e.g. time and current.): It is difficult to read the current value and check stopwatch at the same time	Take a video of the readings and pause at the respective timings to take readings.

5. How can the experiment be modified to study real life scenarios

- Changes and how these changes are relevant
 - E.g. Use salt water so water condition is similar to sea water

*Tabulate: MUST draw table!

Newtonian mechanics

- For SPRING, see if the length required includes the ends or just the spring itself!

Oscillations

- Time for **at least 20s** & obtain two sets of readings for stopwatch
- Difference in **timing should be < 0.20s** (less than human reaction time)
- READ QUESTION CAREFULLY to see if asking for *total time taken t* OR *period T*

Circuits

- Jockey is **tapped vertically** on the wire to **minimise contact between the jockey and the wire**, hence achieving a **lower uncertainty** for the measurement of the balanced length.
- Jockey is **tapped lightly** on the wire to **avoid compressing the wire** which will **affect its resistance**.
- A potentiometer is **more accurate** than a voltmeter as it **does not draw current from the primary circuit**.

Planning

Equipment

- Specify physical quantity, units and instrument used for measurement.
 - Use double slit to measure wavelength
 - Use kinematics (time taken to travel x distance) to find speed

Procedure

- Diagram
 - MUST **LABEL**
 - Indicate specific arrangements
 - If electric motor / heater / any electrical appliance is needed, MUST SHOW variable **DC POWER SOURCE**
 - Increase power input to vary speed of motor / power of heater
 - “Set up apparatus as shown in diagram”
 - Explain how the apparatus is to be used
 - DO NOT NEED PROCEDURE
- Variables
 - Independent variable
 - Dependent variable
 - Controlled variable
 - How can it be changed
 - How can it be measured + Units
 - How can it be kept constant
- Preliminary readings
 - A statement on what needs to be done to obtain the range which the independent variable can be varied
 - No need for actual values
- Repeat to obtain a total of 6 readings

Data and graph

- Equation
- Linearization
 - Plotting _____ against _____, if a straight line (that does / does NOT pass through the origin) is obtained, where _____ is the gradient, _____ is the y-intercept, then relationship is valid.
- State what the gradient and y-intercept mean.

Precautions

*Must be **specific** to the question. No generic answers.

- Safety precautions Accuracy / reliability
 - **Preliminary readings** should be taken to determine the initial velocity and angle of projection for a suitable horizontal range.
 - **Repeat experiment to obtain average**
 - Use **cushion** to prevent glass cylinder from breaking (involving oil)
 - Use **gloves** to **prevent skin irritation** from contact with oil
 - Use **cloth/tray to absorb oil**
 - Use **video frame by frame playback** to determine length/angle ***with a ruler/protractor placed in frame as reference***

- Use **a larger protractor with finer divisions** to **reduce percentage uncertainty**
- Use **spirit level** to **ensure that setup is horizontal**
- Time at least **20 rotations**
- Time at least **20s**
- **Repeat** and **take average** to find period T
- **Allow time between readings** for **experiment to stabilize**
- Use **safety screen/safety goggles** to **prevent objectile from injuring eye**
- Conduct experiment in a **draught free environment** to prevent wind from affective motion of objectile
- Use **high density ball** to **reduce effects of resistive forces**
- Use a **plumbline** for a **more accurate measurement** of trajectory range when the projectile motion is not on a single plane
- Wear **insulating rubber gloves** and bots to **prevent electric shock**
- **Conduct experiment in a dark room** to see light better
- Wear protective glasses to prevent bright light from damaging eyes
- Lamp **becomes hot over time** and can **cause burns when touched**. Switch off when not in use.
- Use **Helmholtz coils** or **large magnets** to produce uniform magnetic field
- Use **Hall probe** to **measure magnetic flux density**
- If experiment is relatively safe, write:
 - The experiment is relatively safe as no high voltage is used / no hot objects are involved / no sharp objects are involved...